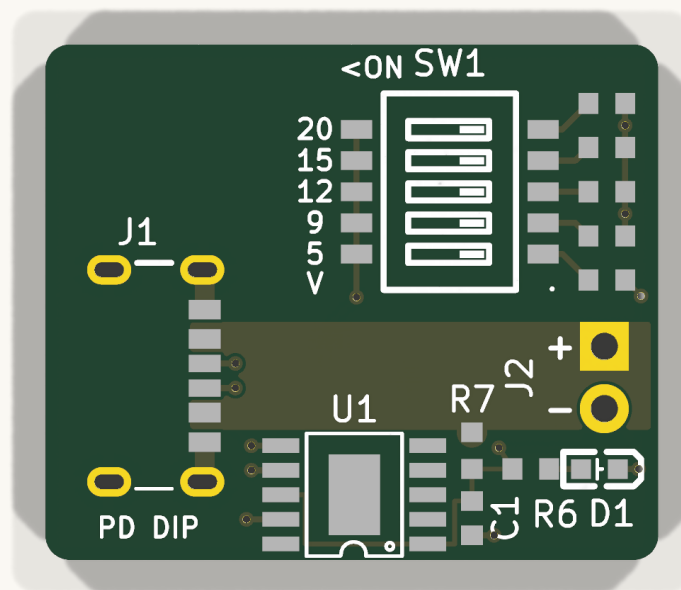


pd-trigger-lite-dip

A 25 by 21 mm USB-C PD trigger: flip one DIP switch to get 5, 9, 12, 15 or 20 V at up to 3 A on two solder pads.

This board asks a USB-C Power Delivery charger for one fixed voltage and passes the charger's VBUS straight to two 2.54 mm output holes. There is no regulator in the power path, so the output is exactly what the charger delivers. You pick the voltage with a five-position DIP switch, one switch per voltage, so changing it needs no soldering iron. It is the pd-trigger-lite board with its solder links replaced by that switch, and 5 V added as a choice.



TOP VIEW 25 X 21 MM, 2 LAYERS, 1 OZ RENDER

USB-C receptacle on the left, the voltage switch SW1 at the top with the voltage printed beside each position, the PD controller at the bottom, output pads on the right (+ is VBUS, - is ground). The red LED sits under the output pads.

Setting the voltage

Exactly one switch ON, set before you plug the charger in.

SWITCH ON	OUTPUT	OTHERS	WHAT IT DOES
1	5 V	all OFF	R5 100k pulls CFG1 up to VHV
2	9 V	all OFF	R1 6.8k from CFG1 to ground
3	12 V	all OFF	R2 24k from CFG1 to ground
4	15 V	all OFF	R3 56k from CFG1 to ground
5	20 V	all OFF	R4 120k from CFG1 to ground
none	undefined		CFG1 floats; do not use
two or more	not allowed		can damage U1 (see below)

On the board the voltage is printed beside each switch: 20, 15, 12, 9 and 5 from the top, so position 1 (5 V) is the bottom one, nearest the output pads. A switch is ON when its slider is pushed toward the side marked ON on the switch body. The board's silkscreen shows that side as "<ON" (toward the voltage labels), but the mark printed on the switch itself is the one to trust. Use a toothpick or a small screwdriver; the sliders are 1.27 mm apart.

EXACTLY ONE SWITCH, SET BEFORE PLUGGING IN

The controller reads the switch once, when it powers up, so set it with the board unplugged and re-plug after any change. With every switch OFF, CFG1 floats and the request is undefined. Never turn on two at once: with switch 1 and any other ON, R5 and the ground resistor form a divider from VHV, and once VHV rises above 5 V that can push CFG1 past its 3.8 V absolute maximum.

Output. The two plated holes take 20 AWG wire or a 1x2 2.54 mm header, which you solder yourself; JLC does not fit them. The rating is 3 A at any voltage. A charger without an e-marked cable never offers more than 3 A, and the board makes no 5 A claim.

LED. D1 is lit whenever VBUS is present. Its current rises with the voltage, so it glows faintly at 5 V and clearly at 12 V and above.

First power-up. Plug in with nothing on the output, measure the voltage across the pads, and only then connect the load. The charger must offer the chosen voltage as a fixed PDO; many phone chargers stop at 9 V, and a charger that lacks it leaves the output at 5 V. The board has no fuse and no TVS diode, because the controller's datasheet asks for neither. A hot-plug spike at 20 V is the one plausible way to damage it, so at 20 V plug the charger into

the wall first and the cable into the board last.

What is on it

Twelve parts placed by JLC, three of them Extended. Everything else is bare copper.

REF	PART	LCSC	TYPE	QTY
U1	CH224A PD sink controller, ESSOP-10	C42459160	Extended	1
J1	USB-C receptacle 6P, 3 A 20 V (TYPE-C-6M-001)	C2798175	Extended	1
SW1	DIP switch, 5 positions, 1.27 mm SMD (SHOU HAN 1.27-5P TPPT)	C7421518	Extended	1
C1	1 uF 50 V X5R 0603	C15849	Basic	1
R1	6.8k 1% 0603 (9 V)	C23212	Basic	1
R2	24k 1% 0603 (12 V)	C23352	Basic	1
R3	56k 1% 0603 (15 V)	C23206	Basic	1
R4	120k 1% 0603 (20 V)	C25808	Basic	1
R5	100k 1% 0603 (5 V)	C25803	Basic	1
R6	10k 1% 0603	C25804	Basic	1
R7	0 ohm 0603	C21189	Basic	1
D1	red LED 0603	C2286	Basic	1

Not assembled: J2, the pair of output holes. The three Extended parts each add JLC's per-part feeder fee; no Basic part can replace any of them.

What it costs

About 33 USD for five assembled boards before shipping, as an estimate.

27.42	5.44	32.9
USD, JLC estimate for 5 boards: PCB 2.00, setup 8.00, 3 Extended fees 9.00, stencil 8.00, joints 0.42.	USD, parts for 5 boards at LCSC unit prices, about 1.09 per board (the switch is 0.53).	USD total for 5 boards, before shipping and tax.

These are estimates from the skill's price table (verified 2026-07-24), not a quote. Live JLC prices have run 1.9 to 3.1 times higher than that table for bare PCBs, so expect the PCB line to be nearer 4 to 6 USD. Shipping depends on the country and the carrier and is not included. The cart at <https://cart.jlcpcb.com/quote> is the only real price.

Estimate from fab/quote.json (order_quote.py, qty 5, HASL, economic assembly) and kicad/parts.json prices, read 2026-09-24.

Ordering at JLCPCB

Three files go up: the gerber zip, then BOM.csv and CPL.csv for assembly.

1. Open the JLC quote page (link above) and upload the gerber zip, fab/pd-trigger-lite-dip_gerbers.zip.
2. Check that the viewer reads 2 layers, 25 x 21 mm. Keep 1.6 mm, 1 oz, HASL, green, quantity 5.
3. Turn on PCB Assembly: Economic, top side, 5 boards.
4. Upload fab/BOM.csv and fab/CPL.csv. Confirm all twelve parts matched their LCSC numbers and that J2 is absent, since it is not assembled.
5. In the placement preview, check four parts. SW1: its pin-1 mark sits at the bottom-right corner, next to R5, so switch 1 is the 5 V row; if JLC's model shows the ON side, it should face the voltage labels. D1: its cathode mark must face the right board edge. U1: the pin-1 dot sits at the corner nearest the capacitor C1. J1: the receptacle mouth points off the left edge. No rotation correction was applied to the CPL, because every footprint comes from JLC's own library.
6. Run JLCDFM on the gerbers as a second opinion, then pay. Payment is yours to do.

Gates on 2026-09-24: ERC 0/0, DRC 0 errors 0 unconnected, place, verify and DFM pass.